

NETELIXIR AIGNITION 3.0 · TEAM TECHBLAZERS

Multi-Channel Ad Revenue Forecasting

Probabilistic revenue forecasts, budget optimization, and grounded AI narratives — a walkthrough of every screen in the app, what it shows, and why it's built that way.

Interface Walkthrough

What This System Does

Ingests

Google Ads, Meta Ads, and Bing Ads campaign data — with automatic schema mapping and mangled-file tolerance.

Optimizes

Where today's existing budget should move, using fitted Hill saturation curves per channel/campaign type.

Explains

A grounded AI narrative layer that only states numbers it can verify against the model's own output.

Forecasts

30/60/90-day revenue as a probability range (P10–P90), not a single fragile number.

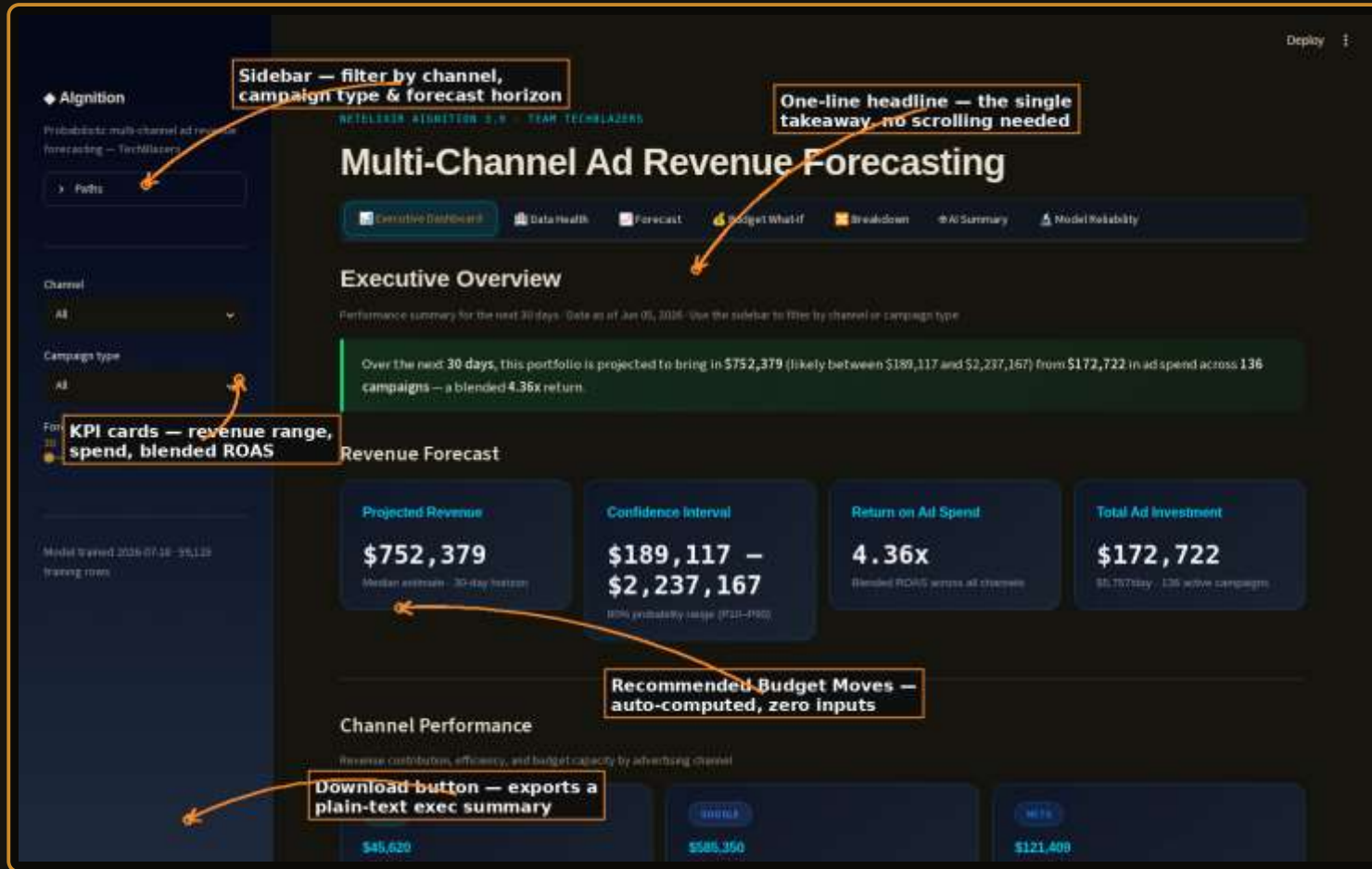
Reconciles

Campaign → campaign type → channel → total, so every rollup sums exactly, enforced by MinTrace.

Proves itself

Backtests, calibration diagrams, and naive-baseline comparisons — the model shows its own homework.

01 Executive Dashboard



WHAT IT SHOWS

The one screen a non-technical stakeholder needs: a plain-English headline, KPI cards, per-channel recommendations, auto-computed budget moves, and a downloadable summary.

WHY IT'S BUILT THIS WAY

- Leads with a single sentence — projected revenue, range, spend, and ROAS — before any chart, so the takeaway never requires scrolling.
- Recommended Budget Moves reuses the same Hill-curve allocator as Budget What-If, but needs zero inputs — it defaults to today's existing spend.
- Every channel always gets a recommendation card, even a 'Stable — keep monitoring' one, so nothing silently drops off the page.
- A 'Download summary (.txt)' button lets the reader forward or print the takeaway without screenshotting the dashboard.

02 Data Health



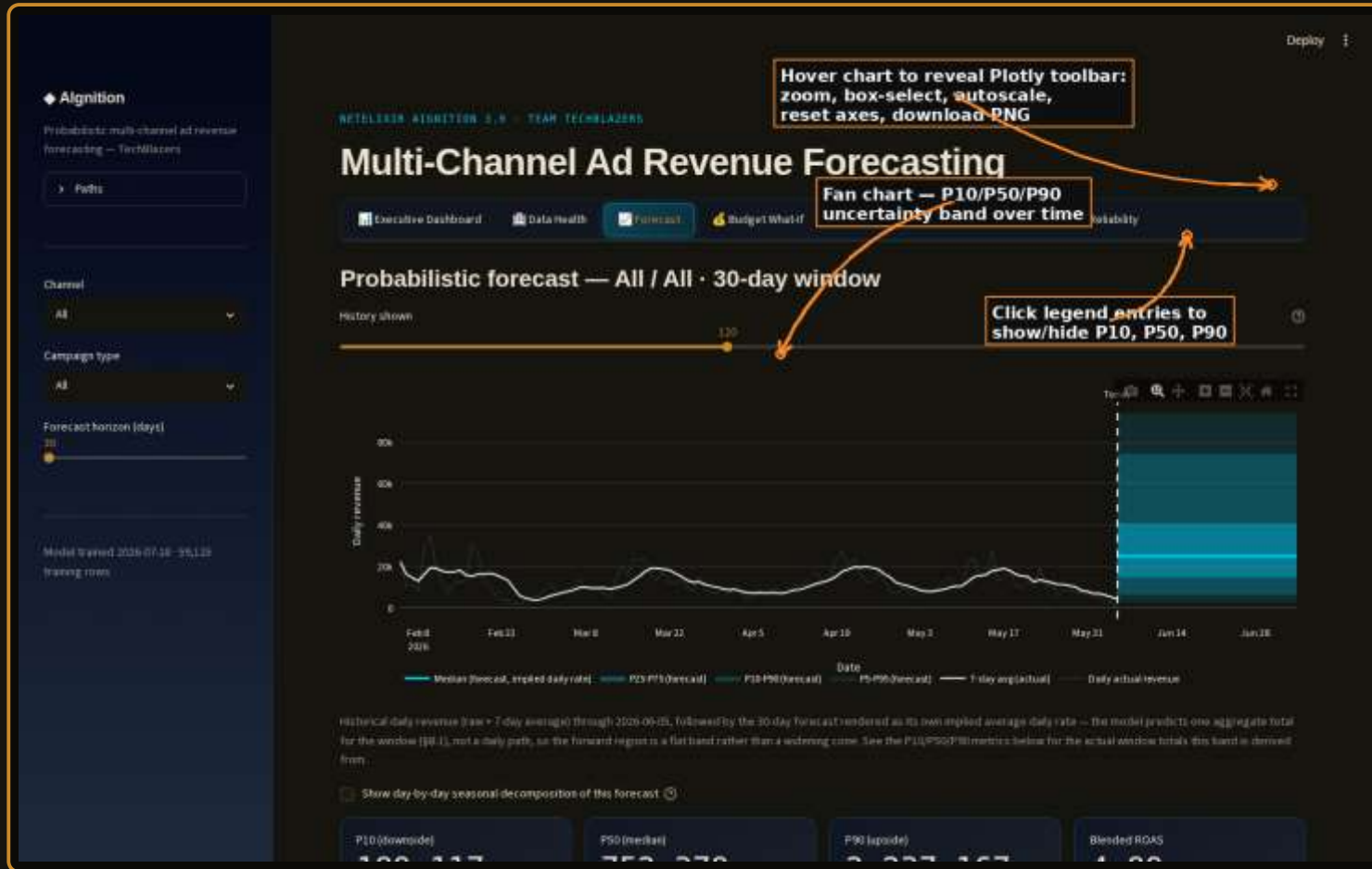
WHAT IT SHOWS

A live, per-file ingestion report — recomputed on every run from whatever is in --data-dir, not cached from training time.

WHY IT'S BUILT THIS WAY

- Shows exactly how each column in a raw export was mapped (exact vs. fuzzy match), what was ignored, and what was left as genuine missing data rather than fabricated.
- Flags revenue confidence per channel — direct vs. proxy (conversion-value-derived) — so a viewer knows which numbers are measured vs. estimated.
- Built to survive real-world messiness: mangled headers, extra columns, and missing fields don't break the pipeline, they get reported.

03 Forecast



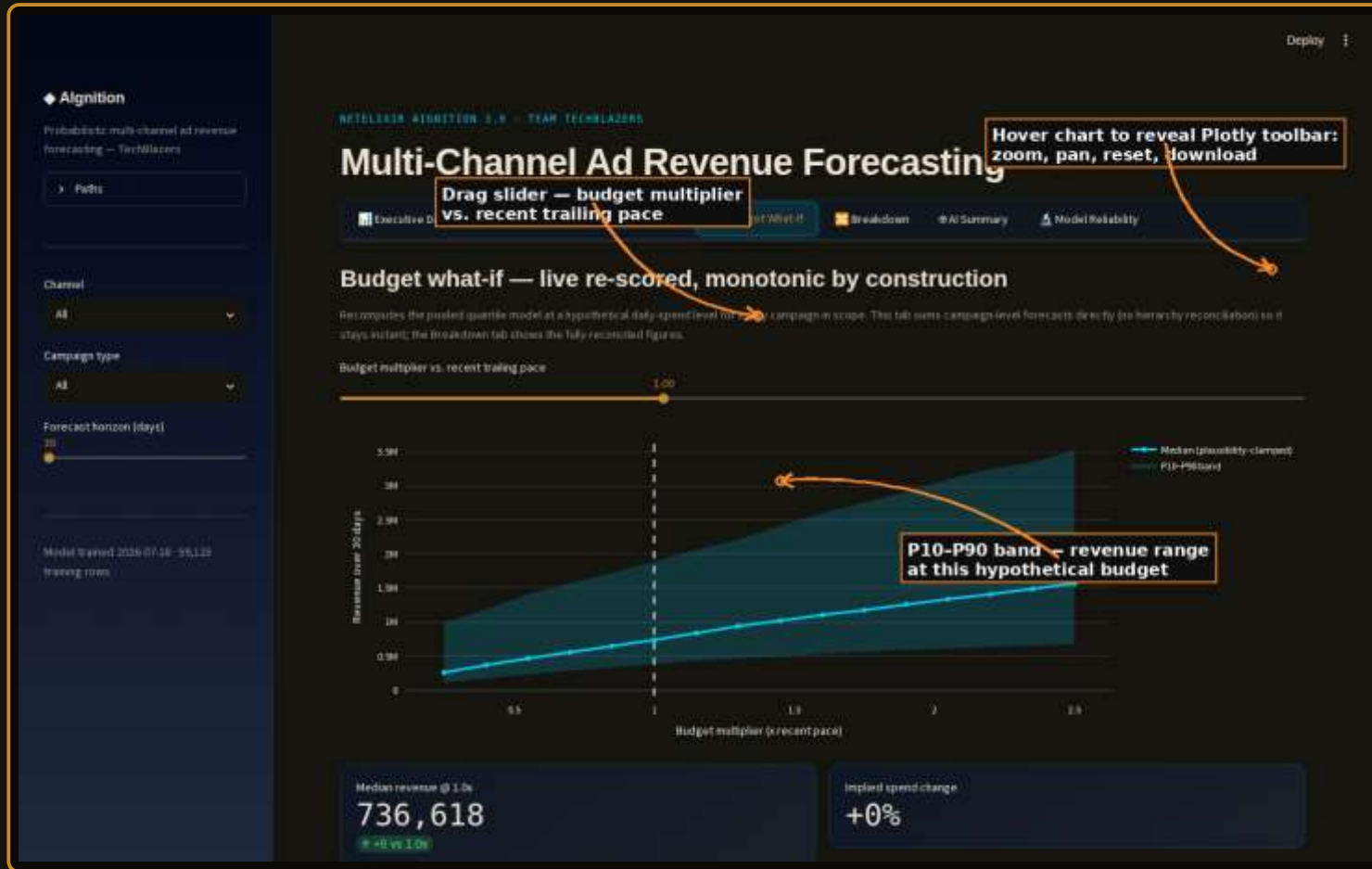
WHAT IT SHOWS

The core probabilistic forecast: historical daily revenue plus a forward-looking P10–P90 fan chart for the selected horizon and scope.

WHY IT'S BUILT THIS WAY

- The forecast is a range, not a point estimate — pretending to know the exact number would be dishonest for a real budget decision.
- Hover any point on the chart for the exact percentile value; hover the chart itself to reveal Plotly's native toolbar — zoom, box-select, pan, autoscale, reset axes, and PNG export.
- An optional day-by-day seasonal decomposition view breaks the flat forecast window into an implied daily shape, for teams that want a calendar-level view.

04 Budget What-If



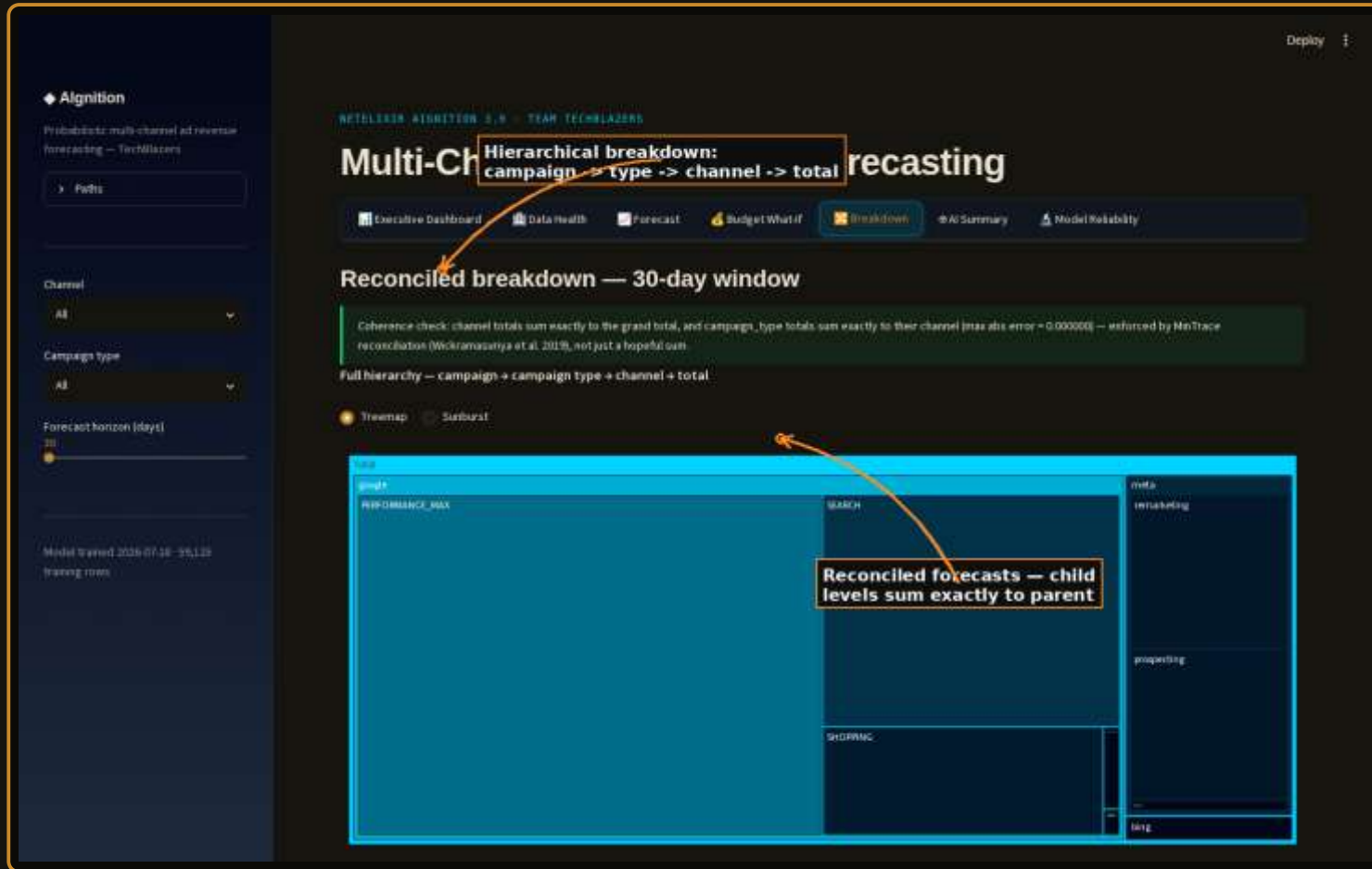
WHAT IT SHOWS

A live, instantly-recomputed 'what if I spent 1.5x' slider — re-scores the model at any hypothetical daily-budget multiplier.

WHY IT'S BUILT THIS WAY

- Monotonic by construction: dragging the slider up never produces a lower forecast than a lower multiplier — enforced numerically, not just visually.
- Sums campaign-level forecasts directly (no hierarchy reconciliation) specifically to stay instant under a slider drag; the Breakdown tab has the fully reconciled version for final numbers.
- The Hill-curve explorer below shows the fitted response curve per channel/campaign type, with the current operating point marked — this is what powers the Executive Dashboard's automatic recommendations.

05 Breakdown



WHAT IT SHOWS

The fully reconciled hierarchy — campaign → campaign type → channel → total — as an interactive treemap/sunburst plus per-level bar charts with uncertainty bands.

WHY IT'S BUILT THIS WAY

- Coherence is enforced, not hoped for: channel totals sum exactly to the grand total (MinTrace reconciliation, Wickramasuriya et al. 2019) — verified every run, max error printed on-screen.
- Treemap/sunburst communicates proportions at a glance; the bar charts underneath add the P10–P90 band a treemap can't show.
- Click any segment to drill in — starts collapsed at campaign-type level since 168 individual campaigns is too dense to show flat.

06 AI Summary

Multi-Channel AI-generated narrative — every number is verified against model output

Grounded AI causal summary
Rule-based narrator (deterministic, always grounded)

For the overall account over the next 30 days, the median forecast is 752,379 in revenue, with a [80,189,117, \$2,237,167]. Revenue is down 8.9% vs. the prior comparable period. Budget saturation status: mixed across channels.

Key drivers

- planned_future_daily_budget (importance rank 1)
- horizon_days (importance rank 2)
- campaign_expansion (importance rank 3)
- revenue_roi_std_28 (importance rank 4)
- trailing_roas_56 (importance rank 5)

Risk flags

- Anomaly flagged on 2026-03-31: revenue z-score 5.1

Source label: LLM-generated vs. rule-based fallback (no API key needed)

This is a rule-based narration of the grounding context (no LLM call available in this run) — every number above comes directly from the deterministic stats engine, not a generative model.

Grounding context (the only facts the narrator was allowed to use)

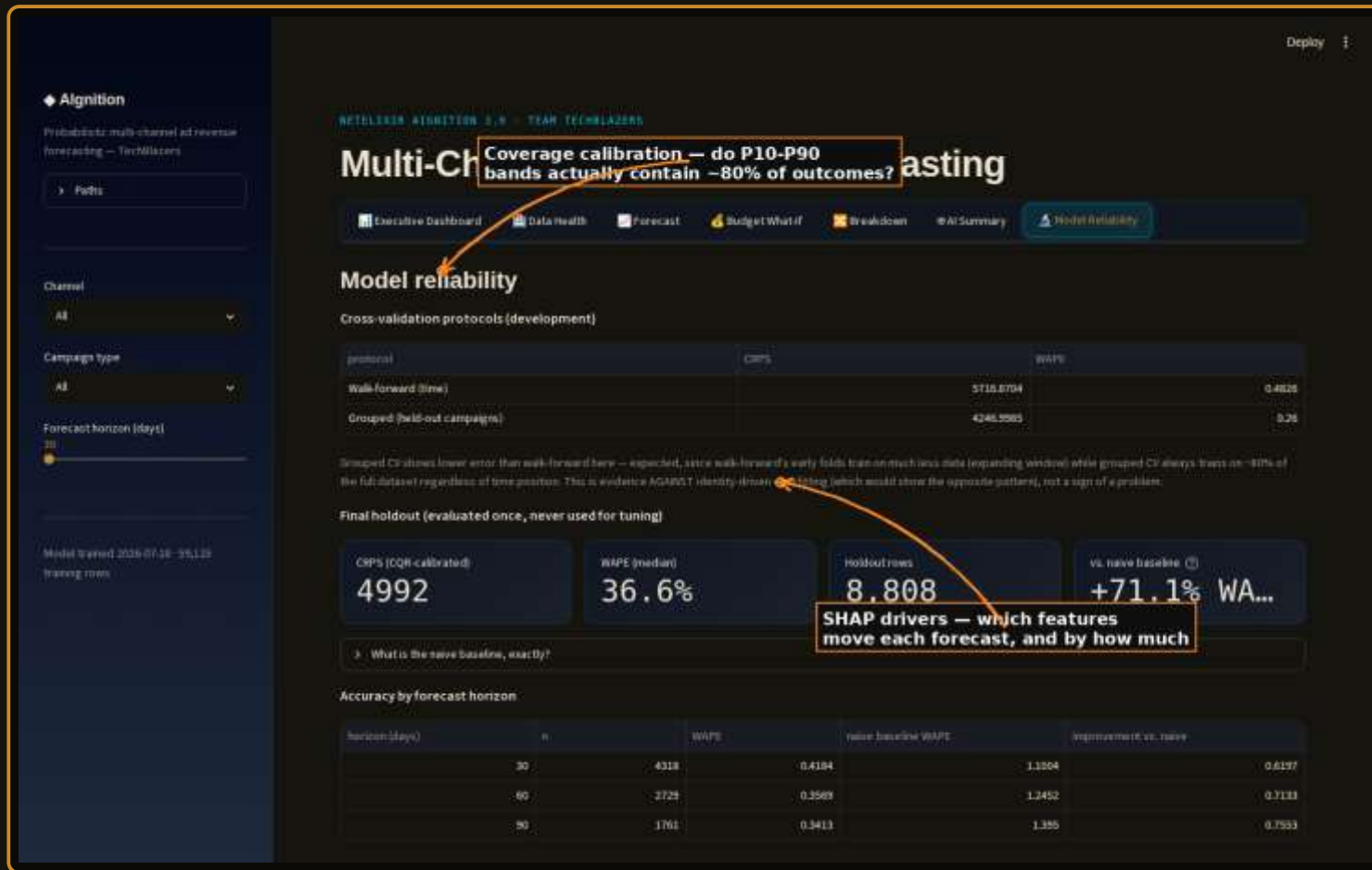
WHAT IT SHOWS

A narrative summary of the forecast, written either by an LLM or by a deterministic rule-based fallback — every number is checked against the model's own output before it's shown.

WHY IT'S BUILT THIS WAY

- Grounded by construction: the narrator is only allowed to state facts from a pre-computed 'grounding context' — it cannot invent a number that isn't already verified.
- Works with zero internet access or API key: falls back to a rule-based narrator using the same grounding data, so the demo never depends on a live LLM call succeeding.
- Every card is labeled with its source (LLM-generated vs. rule-based) so a reader always knows which kind of summary they're reading.

07 Model Reliability



WHAT IT SHOWS

The model's own report card: cross-validation protocols, holdout accuracy, naive-baseline comparison, and a calibration diagram checking whether the P10–P90 bands are honest.

WHY IT'S BUILT THIS WAY

- Reliability diagram plots nominal vs. empirical coverage — if the 80% band doesn't actually contain ~80% of outcomes, this is where that would show up.
- Compares against a naive baseline at every horizon, so 'the model works' is a relative, falsifiable claim, not an assertion.
- A held-out test set, evaluated once and never used for tuning, keeps the headline accuracy numbers honest.

Built to Survive Scrutiny

- ▶ Runs clean on a fresh git clone, fresh venv, zero manual fixes — verified end-to-end.
- ▶ No internet dependency at inference time: LLM narration degrades gracefully to a grounded rule-based fallback.
- ▶ Every headline number is traceable back to a specific, testable computation — 172+ passing tests.
- ▶ Uncertainty is never hidden: every forecast ships as a range, every allocator decision cites the marginal ROAS behind it.